

## SAFETY DATA SHEET

Revision Date 24-Dec-2021

Revision Number 5

### 1. Identification

**Product Name** 2-Acetyl-5-norbornene, tech. mixture of endo and exo

**Cat No. :** AC146870000; AC146870050

**Synonyms** Methyl 5-norbornen-2-yl ketone; 5-Acetylbicyclo[2.2.1]hept-2-ene

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Flammable liquids

Category 4

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Combustible liquid

##### **Precautionary Statements**

###### **Prevention**

Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Wear protective gloves/protective clothing/eye protection/face protection

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

**Other hazards**

Stench.

### 3. Composition/Information on Ingredients

| Component                                   | CAS No    | Weight % |
|---------------------------------------------|-----------|----------|
| Methyl (8,9,10-trinorborn-5-en-2-yl) ketone | 5063-03-6 | 100      |

### 4. First-aid measures

|                                            |                                                                                                                              |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.              |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention. |
| <b>Inhalation</b>                          | Remove from exposure, lie down. Remove to fresh air.                                                                         |
| <b>Ingestion</b>                           | Clean mouth with water. Get medical attention.                                                                               |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                 |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                        |

### 5. Fire-fighting measures

|                                         |                                                                                                                                |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                       |
| <b>Flash Point</b>                      | 61 °C / 141.8 °F                                                                                                               |
| <b>Method -</b>                         | No information available                                                                                                       |
| <b>Autoignition Temperature</b>         | No information available                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                |
| <b>Upper</b>                            | No data available                                                                                                              |
| <b>Lower</b>                            | No data available                                                                                                              |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                       |

**Specific Hazards Arising from the Chemical**

Combustible material. Combustible material. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
0

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Remove all sources of ignition. Take precautionary measures against static discharges.                                                          |
| <b>Environmental Precautions</b>            | See Section 12 for additional Ecological Information.                                                                                           |
| <b>Methods for Containment and Clean Up</b> | Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Sweep up and shovel into suitable containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Avoid contact with skin and eyes. Do not breathe mist/vapors/spray. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition.                                                              |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents. Strong bases. Strong reducing agents. |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                             |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                               |
| <b>Engineering Measures</b>          | None under normal use conditions. Ensure adequate ventilation, especially in confined areas.                                                                                |
| <b>Personal Protective Equipment</b> |                                                                                                                                                                             |
| <b>Eye/face Protection</b>           | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166. |
| <b>Skin and body protection</b>      | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                   |
| <b>Respiratory Protection</b>        | No protective equipment is needed under normal use conditions.                                                                                                              |
| <b>Hygiene Measures</b>              | Handle in accordance with good industrial hygiene and safety practice.                                                                                                      |

## 9. Physical and chemical properties

|                                         |                                       |
|-----------------------------------------|---------------------------------------|
| <b>Physical State</b>                   | Liquid                                |
| <b>Appearance</b>                       | Dark yellow                           |
| <b>Odor</b>                             | Stench                                |
| <b>Odor Threshold</b>                   | No information available              |
| <b>pH</b>                               | No information available              |
| <b>Melting Point/Range</b>              | No data available                     |
| <b>Boiling Point/Range</b>              | 84 - 85 °C / 183.2 - 185 °F @ 18 mmHg |
| <b>Flash Point</b>                      | 61 °C / 141.8 °F                      |
| <b>Evaporation Rate</b>                 | No information available              |
| <b>Flammability (solid,gas)</b>         | Not applicable                        |
| <b>Flammability or explosive limits</b> |                                       |

|                                        |                                  |
|----------------------------------------|----------------------------------|
| Upper                                  | No data available                |
| Lower                                  | No data available                |
| Vapor Pressure                         | No information available         |
| Vapor Density                          | No information available         |
| Specific Gravity                       | 1.005                            |
| Solubility                             | No information available         |
| Partition coefficient; n-octanol/water | No data available                |
| Autoignition Temperature               | No information available         |
| Decomposition Temperature              | No information available         |
| Viscosity                              | No information available         |
| Molecular Formula                      | C <sub>9</sub> H <sub>12</sub> O |
| Molecular Weight                       | 136.19                           |

## 10. Stability and reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                               |
| Stability                        | Stable under normal conditions.                                                          |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. |
| Incompatible Materials           | Strong oxidizing agents, Strong bases, Strong reducing agents                            |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**  
**Toxicologically Synergistic** No information available

### Products

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                           | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------------------------|-----------|------------|------------|------------|------------|------------|
| Methyl<br>(8,9,10-trinorborn-5-en-<br>-2-yl) ketone | 5063-03-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** COMBUSTIBLE LIQUID, NOT REGULATED FOR TRANSPORT IN THIS QUANTITY  
According to 49 CFR §173.150(f)(1), this material should be reclassified as NA1993, Combustible Liquid, NOS if it is shipped in bulk.

**UN-No** NA1993  
**Proper Shipping Name** Combustible liquid, n.o.s.  
**Packing Group** III

**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### United States of America Inventory

| Component                                         | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------------------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Methyl<br>(8,9,10-trinorborn-5-en-2-yl)<br>ketone | 5063-03-6 | -    | -                                             | -                           |

### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA 12(b)** - Notices of Export Not applicable

### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL |
|-----------|-----------|-----|------|-----------|-------|------|------|------|-------|------|
| Methyl    | 5063-03-6 | -   | -    | 225-767-0 | -     | -    | X    | -    | -     | -    |

|                                         |  |  |  |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|--|--|--|
| (8,9,10-trinorborn-5-en-2-yl)<br>ketone |  |  |  |  |  |  |  |  |  |  |
|-----------------------------------------|--|--|--|--|--|--|--|--|--|--|

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

**SARA 313** Not applicable

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA** Not applicable

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations** Not applicable

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

### **Authorisation/Restrictions according to EU REACH**

### **Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                         | CAS No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Methyl<br>(8,9,10-trinorborn-5-en-2-yl)<br>ketone | 5063-03-6 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                         | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Methyl<br>(8,9,10-trinorborn-5-en-2-yl)<br>ketone | 5063-03-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

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## 16. Other information

|                  |                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared By      | Regulatory Affairs<br>Thermo Fisher Scientific<br>Email: EMSDS.RA@thermofisher.com                                                                                                                                                      |
| Revision Date    | 24-Dec-2021                                                                                                                                                                                                                             |
| Print Date       | 24-Dec-2021                                                                                                                                                                                                                             |
| Revision Summary | This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS). |

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**